

Paper B-11: Computational Architecture: A Constraint-Driven Flux Dynamics (CDFS) Perspective

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Abstract

In this paper we apply Adaptive Flux Limitation (AFL) to computer architecture, modelling instruction throughput as constrained transport governed by the potential Φ (instruction retirement rate, FLOP/s), with multi-dimensional constraint $C = (C_{mem}, C_{pwr}, C_{IPC})$ encoding memory bandwidth, thermal power budget, and instruction-level parallelism limits. The effective $\Psi_s^* = J/C_{eff}$ where $C_{eff} = \min(C_{mem}, C_{pwr}, C_{IPC})$ classifies processor states: memory-bound, power-bound, or IPC-bound.

We model the hypothesis that roofline performance models, cache hierarchy design, heterogeneous chip architecture (CPUs + GPUs + NPUs), and dark-silicon management emerge naturally from AFL multi-wall dynamics. Dennard scaling failure is recast as C_{pwr} wall formation; modern chiplet design is AFL constraint diversification.

Status: Inferred from AFL first principles; quantitative predictions are hypothesis-level.

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Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Modern processors contain $> 10^{10}$ transistors operating at $> 10^9$ Hz, yet practical throughput is limited by three architectural constraints that have converged to form permanent performance ceilings. These “walls” are among the most consequential engineering challenges in computer science:

1. **Memory wall:** DRAM bandwidth grows at $\sim 26\%$ /year while CPU performance grew at $\sim 52\%$ /year (through 2004), creating an ever-widening gap;
2. **Power wall:** Dennard scaling broke down around 2005; frequency can no longer increase without thermal runaway (“dark silicon”);
3. **IPC wall:** instruction-level parallelism (ILP) extractable by out-of-order execution saturates at $\sim 4\text{--}6$ IPC for most workloads.

AFL unifies these three walls as a multi-dimensional constraint system: processor throughput J is jointly bounded by three independent C values, and the effective constraint is $C_{eff} = \min(C_{mem}, C_{pwr}, C_{IPC})$ — the weakest wall determines performance. This provides a single $\Psi_s^* = J/C_{eff}$ diagnostic for architectural bottleneck ($\Psi_s > 1$ lock state) identification.

3 Core Hypothesis

Computational throughput in processor architecture is governed by a multi-dimensional AFL system in which three converging constraints (memory bandwidth, power budget, instruction parallelism) jointly bound instruction retirement rate, with heterogeneous chip design representing the AFL strategy for constraint diversification across varied workload types.

AFL architecture hypothesis: the roofline performance model is an AFL two-constraint model (C_{mem} and $C_{peak-FLOP}$), and the “roofline knee” is the AFL $\Psi_s^* = 1$ operating point where compute and memory constraints are balanced. Modern performance engineering seeks workloads where $\Psi_s^* \approx 1$ on the compute roofline rather than the memory roofline. **Status: Hypothesis; roofline model provides direct structural correspondence.**

4 Computational Potential Field

We define the generalised processor potential:

$$\Phi = \text{instruction retirement rate} = \text{IPC} \times f \quad (1)$$

where IPC is instructions per cycle and f is clock frequency. The AFL transport equation governs how Φ flows through the processor pipeline:

$$J = -\frac{1}{C_{eff}} \nabla \Phi \quad (2)$$

where C_{eff} encodes the binding constraint at any given workload.

5 The Three-Wall AFL System

5.1 Memory Wall: C_{mem}

Memory bandwidth constraint:

$$C_{mem} = \frac{\text{Arithmetic Intensity}}{\text{Peak Memory Bandwidth}} = \frac{FLOP/byte}{BW_{peak}}. \quad (3)$$

A workload is memory-bound when $J_{compute} > C_{mem}$: the processor can perform more operations than the memory subsystem can supply data for. AFL prediction: memory-bound workloads have $\Psi_{s,mem} > 1$; they waste compute cycles waiting for data.

Cache hierarchy is AFL constraint reduction: L1 cache reduces effective C_{mem} for data with high temporal locality by a factor of ~ 100 (DRAM vs. L1 latency ratio).

5.2 Power Wall: C_{pwr}

Thermal power constraint:

$$C_{pwr} = \frac{P_{TDP}}{P_{per-transistor}} = \frac{P_{TDP}}{\alpha C V^2 f}, \quad (4)$$

where P_{TDP} is thermal design power, C (here capacitance) is switching capacitance per transistor, V is supply voltage, f is frequency.

Dennard scaling: through 2004, $V \propto f$ kept $P_{per-transistor} \propto f^{-2}$, so scaling f preserved P_{total} . When V could not decrease further (leakage current), $P_{pwr} \propto f^3$ and scaling ceased. AFL interpretation: C_{pwr} wall formed when Dennard scaling exhausted the adaptive mechanism ($\beta_{Dennard} \rightarrow 0$).

Dark silicon: when only a fraction f_{active} of transistors can be active simultaneously at design power, $J_{compute} = f_{active} \cdot J_{peak}$. AFL $\Psi_{s,pwr} = J_{peak} / (f_{active} C_{pwr})$: dark silicon is the AFL response to power-wall constraint — reduce J_{active} to maintain $\Psi_{s,pwr} \leq 1$.

5.3 IPC Wall: C_{IPC}

Instruction-level parallelism constraint:

$$C_{IPC} = \frac{1}{ILP_{extractable}} \quad (5)$$

where $ILP_{extractable}$ is the average instruction-level parallelism available in the instruction stream. Out-of-order execution, branch prediction, and speculative execution increase $ILP_{extractable}$, reducing C_{IPC} .

Amdahl's law at the instruction level: sequential data dependencies (RAW hazards) impose irreducible C_{IPC} . Modern OOO processors extract ~ 4 – 6 IPC for integer workloads; the IPC wall is the AFL sequential-fraction constraint at microarchitectural scale.

6 Conservation Dynamics

Processor throughput obeys conservation:

$$\frac{\partial \Phi}{\partial t} + \nabla \cdot J = S \quad (6)$$

where S represents instruction supply (from the instruction cache and decoder) and J represents instruction retirement (completed execution). The pipeline is the AFL transport medium; the reorder buffer (ROB) is the AFL queue.

7 Adaptive Constraint Dynamics

Architectural constraints evolve dynamically:

$$\frac{dC_{eff}}{dt} = \alpha |J| - \beta C_{eff} \quad (7)$$

where α governs workload-induced constraint stress (high throughput fills ROB, causes branch mis-predictions, increases cache pressure) and β governs microarchitectural recovery (branch predictor warms up, prefetcher learns access patterns, TLB reloads).

Physical mechanisms:

- **Branch misprediction:** wrong-path speculation fills ROB; recovery flushes pipeline (transient $J \rightarrow 0$, C spikes);
- **Cache miss cascade:** L1 miss $\rightarrow$ L2 miss $\rightarrow$ L3 miss $\rightarrow$ DRAM: each level increases effective C_{mem} by $\sim 10\times$;
- **TLB miss:** virtual-to-physical address translation miss requires page-walk, adding ~ 100 cycle penalty to effective C ;
- **Hardware prefetcher:** learns memory access patterns, reducing effective C_{mem} by prefetching data before it is needed (β mechanism).

7.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio Ψ_s^* is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \quad (8)$$

where S (Surface Responsiveness) and M_s (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \quad (9)$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \quad (10)$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

8 Flux–Constraint Number

$$\Psi_s^* = \frac{J}{C_{eff}} = \frac{J}{\min(C_{mem}, C_{pwr}, C_{IPC})} \quad (11)$$

8.1 Memory-Bound Regime (C_{mem} dominates)

Workload has low arithmetic intensity; processor waits for data. Memory-bound workloads: streaming operations (memcpy, matrix transpose), sparse linear algebra, large graph traversal. AFL $\Psi_{s,mem} > 1$: processors can compute faster than memory can supply.

8.2 Compute-Bound Regime (C_{IPC} or C_{pwr} dominates)

Workload has high arithmetic intensity; memory latency is hidden. Dense matrix multiplication (GEMM), FFT on cached data, neural network forward pass on GPU. AFL $\Psi_{s,mem} < 1$: memory is not the bottleneck ($\Psi_s > 1$ lock state); compute is the limit.

8.3 Thermally Limited Regime (C_{pwr} dominates)

All transistors operational but power budget exhausted; frequency must be reduced below rated speed. Mobile processors and sustained HPC workloads operate in this regime. AFL $\Psi_{s,pwr} \rightarrow 1$: processor executes at thermal steady-state.

9 The Roofline Model as Two-Wall AFL

The Roofline performance model (Williams, Waterman, Patterson 2009) plots:

$$\text{Attainable GFLOP/s} = \min(P_{peak}, B_{peak} \times I) \quad (12)$$

where I is arithmetic intensity (FLOP/byte), P_{peak} is peak compute, B_{peak} is peak bandwidth. This is the AFL two-wall model:

$$J = \min\left(\frac{\Phi}{C_{mem}}, \frac{\Phi}{C_{IPC}}\right) \quad (13)$$

$$\Psi_s^* = \frac{J}{\min(C_{mem}, C_{IPC})} \quad (14)$$

The roofline “knee” at $I_{knee} = P_{peak}/B_{peak}$ is the AFL $\Psi_s^* = 1$ balanced point. Optimising code for the roofline model is AFL constraint balancing: raise I (increase compute per byte) until $\Psi_{s,mem} = \Psi_{s,IPC} = 1$.

10 Cache Hierarchy as AFL Constraint Reduction

Cache hierarchy reduces effective C_{mem} :

$$C_{mem,eff} = h_1 C_{L1} + (1 - h_1) h_2 C_{L2} + (1 - h_1)(1 - h_2) C_{DRAM} \quad (15)$$

where h_1, h_2 are L1, L2 hit rates and $C_{L1} \ll C_{L2} \ll C_{DRAM}$.

AFL cache design principle: keep C_{L1} small enough that $\Psi_{s,L1} < 1$ for the temporal-locality working set; C_{L2} for the spatial-locality working set. Cache size is AFL constraint buffer sizing.

11 Heterogeneous Architecture as AFL Constraint Diversification

Modern chips combine multiple processor types:

$$\text{CPU} \rightarrow \text{low-latency sequential workloads} \quad (16)$$

$$\text{GPU} \rightarrow \text{high-throughput parallel workloads} \quad (17)$$

$$\text{NPU/TPU} \rightarrow \text{matrix-multiplication workloads} \quad (18)$$

This is AFL constraint diversification: each accelerator has different $(C_{mem}, C_{pwr}, C_{IPC})$, and the heterogeneous system routes workloads to the accelerator where $\Psi_s^* \approx 1$:

$$\text{optimal routing: } J_{workload} \rightarrow \text{accelerator}^* = \arg \min_k \Psi_{s,k} \quad (19)$$

GPUs have $C_{IPC,GPU} \ll C_{IPC,CPU}$ (massively parallel), but $C_{mem,GPU} \sim C_{mem,CPU}$ per core. Matrix operations with high arithmetic intensity ($I > 100$ FLOP/byte) have $\Psi_{s,GPU} \approx 1$ (compute-bound on GPU) while $\Psi_{s,CPU} \gg 1$ (memory-bound on CPU).

12 Chiplet Architecture

Modern chiplet design (AMD EPYC, Apple M-series, Intel Meteor Lake) disaggregates monolithic die into compute tiles, memory tiles, and I/O tiles. This is AFL modularisation:

$$C_{eff,chiplet} < C_{eff,monolithic} \quad (20)$$

because each chiplet is optimised for its constraint type, and inter-chiplet interconnects (UCIe, HBM stacking) are designed for the specific J between chiplet types.

13 Spatial Self-Organization

Processor microarchitecture self-organises into execution hierarchies (pipeline stages), throughput corridors (critical paths), caching hubs (shared L3), and adaptive resource pools (ROB, issue queue). This organisation minimises effective execution constraint while maximising instruction retirement throughput.

AFL-PDE simulations (`supplementary_eng.11.py`) model three-wall AFL: $\Psi_{s,mem}$, $\Psi_{s,pwr}$, $\Psi_{s,IPC}$ evolve under a synthetic workload trace, showing which wall is binding at each time step and how microarchitectural optimisations shift the binding constraint.

14 Predictions

1. **Roofline knee:** optimal arithmetic intensity $I^* = P_{peak}/B_{peak}$; AFL confirms this is the $\Psi_{s,mem} = \Psi_{s,IPC}$ balance point.
2. **Dark silicon fraction:** fraction of active transistors $f_{active} = P_{TDP}/(\alpha_{sw}CV^2fN)$; AFL predicts f_{active} scales inversely with frequency.
3. **Cache sizing:** optimal L1 cache size equals temporal working-set size; AFL predicts $h_1 = 1 - C_{L2}/C_{DRAM}$ at optimal sizing.
4. **Heterogeneous routing:** workloads with $I > I_{GPU}^*$ should route to GPU; AFL predicts routing threshold from accelerator roofline parameters.
5. **Chiplet benefit:** chiplet disaggregation reduces C_{eff} by $\delta C \propto N_{tiles} \cdot \Delta C_{specialisation}$; AFL quantifies chiplet advantage.

Status: Predictions 1–2 consistent with established computer architecture results; predictions 3–5 are AFL extensions requiring microarchitectural measurement.

15 Computational Interpretation

The companion script `supplementary_eng.11.py` implements AFL three-wall architecture:

- workload trace with varying arithmetic intensity $I(t)$;
- three constraint channels: $C_{mem}(I, BW)$, $C_{pwr}(f, V, N)$, $C_{IPC}(ILP)$;
- $\Psi_{s,eff} = J / \min(C_{mem}, C_{pwr}, C_{IPC})$ plotted vs. time;
- roofline: performance vs. intensity curve with memory and compute bounds;
- outputs: three-wall Ψ_s^* timeline, roofline plot, cache sizing recommendation.

16 Discussion

AFL unifies computer architecture under a three-wall multi-constraint framework. The memory wall, power wall, and IPC wall are three independent AFL constraints; the roofline model is two-wall AFL; heterogeneous architecture is AFL constraint diversification; chiplets are AFL modularisation. The AFL architectural design principle: minimise $C_{eff} = \min(C_{mem}, C_{pwr}, C_{IPC})$ by balancing across all three constraints simultaneously.

The key AFL insight: architectural optimisation should target $\Psi_{s,mem} = \Psi_{s,pwr} = \Psi_{s,IPC} = 1$ simultaneously (all walls equally binding), which maximises efficient utilisation of all resources. Modern heterogeneous designs approach this ideal by routing each workload type to the accelerator where its specific wall is the binding constraint.

17 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

17.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

18 Major Universal Upgrade: Mechanism, Scale, and Tests

18.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Computational Architecture, the proposed drive is instructions, data, memory requests, and synchronization. The active limitation is memory bandwidth, dependencies, energy, latency, and physical layout. Responsiveness is carried by parallelism, caching, speculation, scheduling, and specialization, while retained state is represented by architectural state, compiler choices, cache history, and accumulated heat. The outcome to explain is throughput, latency, energy efficiency, and correctness.

Table 1: Operational CDFD/AFL map for Paper B-11.

Term	Paper-specific observable meaning	Measurement question
Φ	instructions, data, memory requests, and synchronization	What rate, load, gradient, or demand enters the focal system?
C	memory bandwidth, dependencies, energy, latency, and physical layout	Which measured bottleneck limits transfer, function, or recovery?
S	parallelism, caching, speculation, scheduling, and specialization	Which process changes effective capacity or reroutes the load?
M_s	architectural state, compiler choices, cache history, and accumulated heat	Which retained state makes equal present loads produce different futures?
Y	throughput, latency, energy efficiency, and correctness	Which outcome is predicted out of sample, with uncertainty?

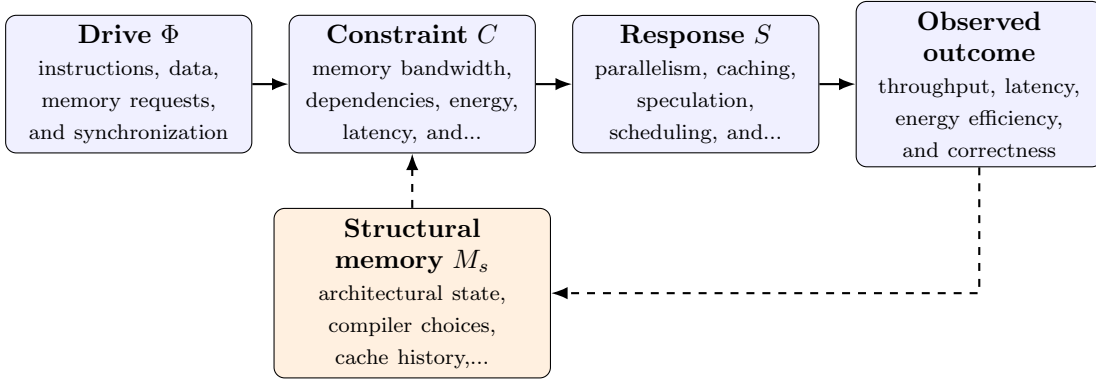


Figure 1: Paper B-11 observable CDFD/AFL architecture for Computational Architecture. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

18.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (21)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (22)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (23)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (24)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in instructions, data, memory requests, and synchronization arrives faster than parallelism, caching, speculation, scheduling, and specialization can compensate. This raises or redistributes memory bandwidth, dependencies, energy, latency, and physical layout. If the load persists, the retained state encoded by architectural state, compiler choices, cache history, and accumulated heat changes the next response, so a later event of equal magnitude can produce a different trajectory in throughput, latency, energy efficiency, and correctness. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

18.3 Cross-Scale Universality Without Mechanistic Erasure

For the engineered systems family, the universal comparison is between demand, designed capacity, feedback control, dependency topology, and accumulated technical state. The strongest engineering use is prospective: the mapping must identify a bottleneck, a control action, or a recovery signature before the failure occurs. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow-constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is picoseconds to hardware generations; transistors to data centers. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

18.4 Evidence Ladder and Discriminating Tests

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase

Table 2: Evidence ladder for Paper B-11. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure hardware counters, traces, benchmarks, thermal data, and architecture simulations and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a memory-intensive workload under power or thermal limits while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: the variables do not improve performance-portability or bottleneck prediction.

estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

18.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from hardware counters, traces, benchmarks, thermal data, and architecture simulations and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of throughput, latency, energy efficiency, and correctness. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

19 Conclusion

Computer architecture evolves through constrained instruction-transport dynamics governed by three converging AFL constraints: memory bandwidth, power budget, and instruction parallelism. AFL provides a unified framework describing the roofline model, dark silicon, cache hierarchy, heterogeneous design, and chiplet architecture within a single multi- C AFL system, connecting computational architecture science to all other Part IV Engineered papers through the universal AFL equation.

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